

References and Additional Information

Polymer-Engineered Silicon Nitride Slot Waveguides for Enhanced On-Chip Stimulated Brillouin Scattering

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Additional Information

This document provides the complete reference list and additional information for the ECIO 2026 poster entitled:

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This work presents a numerical investigation of backward stimulated Brillouin scattering (SBS) in polymer-engineered Si₃N₄ slot waveguides. Four polymer materials, BCB, PMMA, CYTOP, and IP-Dip, were evaluated in both conventional and pedestal-supported slot-waveguide configurations.

Among the investigated materials, BCB yielded the highest SBS gain coefficient of approximately 202 W⁻¹m⁻¹, corresponding to an enhancement of about 2.5× compared with previously reported Si₃N₄ SBS waveguides.

The results demonstrate that polymer engineering offers an effective route for enhancing acoustic confinement and SBS performance in Si₃N₄ integrated photonic platforms.

Software Acknowledgment

The numerical simulations presented in this work were performed using the NumBAT finite-element framework [10] for modelling stimulated Brillouin scattering in integrated photonic waveguides.

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Full Reference List

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